MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER RECTIFIER, GAS

TYPE 3B28

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, half-wave

See figure 1

Mounting position: Any

Weight: 4-ounces (113.4 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Io	Ⓕ tk	F	TA
Unit:	Vac	kv	a	mAdc	sec	Hz	°C
Maximum:	2.63	10.0	1.0	250	---	150	90
Minimum:	2.38	---	---	---	10	---	-75
Maximum:	2.63	5.0	2.0	500	---	500	90
Minimum:	2.38	---	---	---	10 (see note 1)	---	-75
TEST CONDITIONS:	2.5	11	---	---	30	60	---

GENERAL:

Qualification - Required

Ⓕ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1231	Peak emission by voltage drop	-	ib = 5.0 a	0.65	II	etd	---	14.0	v
1301	Filament current	-		0.65	II	If	4.60	5.40	Aac
Ⓕ 1353	Operation of rectifiers	3	Full-wave circuit; Io = 250 mAdc (min)/p; t = 120 sec (min)	0.65	II	---	---	---	---
Ⓕ 3201	Critical anode voltage for conduction	2	Rℓ = 25 ohms	0.65	II	epy	---	40	v
	Quality conformance inspection, part 2								
1036	Bump	-	Angle = 20°	---	---	---	---	---	---
	Quality conformance inspection, part 3								
Ⓕ 1501	Intermittent life	-	Group C; operation of rectifiers; tk = 10 (min); 50 minutes "on", 10 minutes "off"	---	---	cy	600	---	cycles
---	Intermittent life-test end points:								
1231	Peak emission by voltage drop	-		---	---	etd	---	17.0	v
3201	Critical anode voltage for conduction	-		---	---	epy	---	90	v
---	Life test	-	Group D; epx = 5.0 kv; ib = 2 a; Io = 500 mAdc/p; F = 400 ±25 Hz; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Peak emission by voltage drop	-		---	---	etd	---	17	v

NOTES:

1. For reliable operation and longer life, it is recommended that at least 10 seconds of cathode warmup time be allowed.
2. Sine wave ac shall be applied to the anode with pin 4 as the circuit return point. Two tests shall be made; using pin 4 as the reference point, (1) anode voltage shall be in phase with the filament voltage on pin 1, and (2) the anode voltage shall be 180° out of phase with the filament voltage on pin 1.
3. This test shall be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

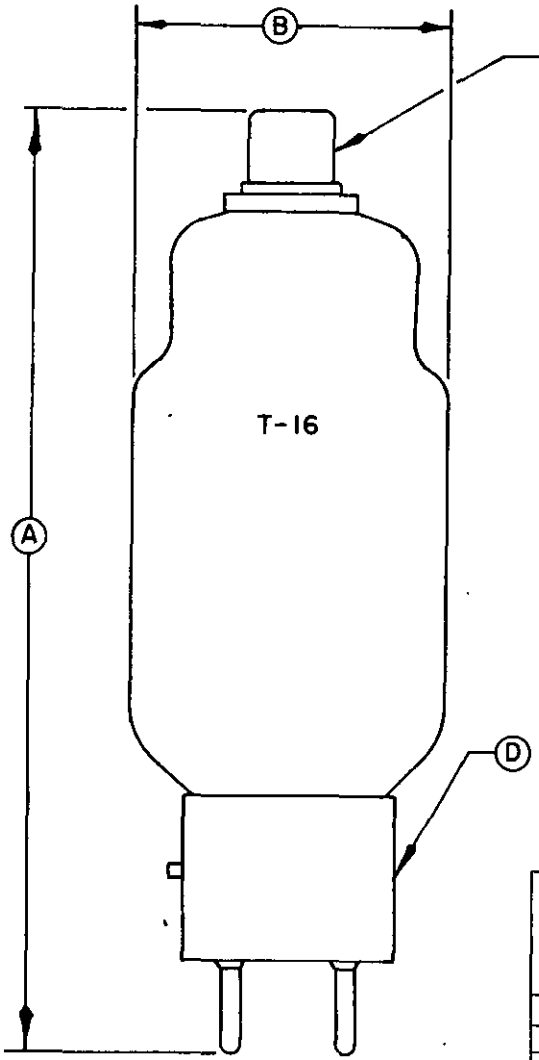
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2934)

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PIN CONNECTIONS	
Pin No.	Element
1	f
2	nc
3	nc
4	f, shield
Cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.87 (149.10)	6.15 (156.21)
B		2.062 (52.37)
(F) Quality conformance inspection, part 3 (see note)		
C	Cap: C1-5 (EIA)	
D	Base: A4-10 (EIA)	

NOTE:
Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 3B28.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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